6th Grade Review — Written Explanation Companion

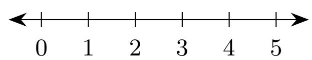
This companion restates each question from the 6th Grade Review, shows the checked answer, and explains the work very slowly. Follow the steps in order. Read every Important words section before you start the numbered steps.

A. Number Lines

A1

Question

Plot the point 4 on the number line below:



Checked answer: A solid dot on the tick mark for 4.

What the question is asking

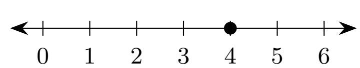
This question asks you to show where the number 4 sits on a straight line of numbers by placing a mark there.

Important words

- A *number line* is a straight line with numbers marked in order from left to right.
- A *tick mark* is a small mark that shows an exact number.
- A *point* on a number line is one exact place. We show it with a solid filled circle (a dot).
- A *whole number* is $0, 1, 2, 3, 4$, and so on.

Work the problem one tiny step at a time

1. Draw a straight horizontal line. Put an arrow on the right end to show the numbers keep going.
2. Mark evenly spaced tick marks. Label them 0, 1, 2, 3, 4, 5, 6 from left to right.
3. Find the tick mark labeled 4.
4. Put a solid filled dot exactly on that tick mark. That means "the number 4 is here."



Final answer: Solid dot at 4 on the number line.

Why this makes sense

The number 4 is a whole number, so it sits on a labeled tick, not between ticks. A common mistake is putting the dot between 3 and 4. The correct place is exactly on the 4 tick.

B. Order Of Operations

Before these problems, learn the order-of-operations rule called **PEMDAS**.

- **P** means *Parentheses*: do work inside ( ) first.
- **E** means *Exponents*: do powers like 2^4 next.
- **MD** means *Multiply* and *Divide*: do these from left to right.
- **AS** means *Add* and *Subtract*: do these from left to right last.

B1

Question
$8 + 3 \times 4 =$
Checked answer: 20

What the question is asking
This question asks for the value of $8 + 3 \times 4$. Do multiply before add.

Important words

- An *expression* is a math phrase with numbers and operations, but no equals sign yet.
- *Multiply* means "times," written $\times$.
- *Add* means "plus," written $+$.
- *PEMDAS* is the order rule: Parentheses, Exponents, Multiply/Divide, Add/Subtract.

Work the problem one tiny step at a time

1. Look at PEMDAS. There are no parentheses and no exponents in this problem.
2. Next comes Multiply/Divide. There is a multiply: 3×4.
3. Compute the multiply first: $3 \times 4 = 12$.
4. Now the expression becomes $8 + 12$.
5. Last comes Add/Subtract. Add: $8 + 12 = 20$.
6. Check: if you wrongly added first ($8 + 3 = 11$, then $11 \times 4 = 44$), that breaks PEMDAS. Multiply comes before add.

Final answer: 20

Why this makes sense
Multiplication comes before addition in PEMDAS. So we turn 3×4 into 12 first, then add 8. That gives 20.

B2

Question

$(21 - 4) \times 3 =$
Checked answer: 51

What the question is asking

This question asks for the value of $(21 - 4) \times 3$. Do the work inside the curved marks first, then multiply.

Important words

- *Parentheses* ( ) are grouping symbols. Work inside them first (the P in PEMDAS).
- *Subtract* means "take away," written $-$.
- *Multiply* means "times," written $\times$.

Work the problem one tiny step at a time

1. PEMDAS says P (Parentheses) comes first.
2. Inside the parentheses: $21 - 4$.
3. Subtract: $21 - 4 = 17$.
4. Now the expression is 17×3.
5. Multiply: 17×3. Break it down: $10 \times 3 = 30$ and $7 \times 3 = 21$, then $30 + 21 = 51$.
6. So $(21 - 4) \times 3 = 51$.

Final answer: 51

Why this makes sense

The parentheses force you to subtract before you multiply. Without parentheses, $21 - 4 \times 3$ would be $21 - 12 = 9$. The parentheses change the order, so the answer is 51.

B3

Question

$2^4 + 1 =$
Checked answer: 17

What the question is asking

This question asks for the value of $2^4 + 1$. First build 2^4 by using 2 as a factor four times, then add 1.

Important words

- An *exponent* (the E in PEMDAS) is a small raised number that tells how many times the base is used as a factor.
- In 2^4, the *base* is 2 and the *exponent* is 4. So 2^4 means $2 \times 2 \times 2 \times 2$.

- A *factor* is a number being multiplied. In $2 \times 2 \times 2 \times 2$, each 2 is a factor.
- *Add* means plus.

Work the problem one tiny step at a time

1. PEMDAS: no parentheses. Next is E (Exponents).
2. Compute 2^4: start with 2.
3. $2 \times 2 = 4$ (that is two factors of 2).
4. $4 \times 2 = 8$ (that is three factors of 2).
5. $8 \times 2 = 16$ (that is four factors of 2). So $2^4 = 16$.
6. Now the expression is $16 + 1$.
7. Add: $16 + 1 = 17$.

Final answer: 17

Why this makes sense

The exponent must be done before the addition. A common mistake is reading 2^4 as $2 \times 4 = 8$. That is wrong. 2^4 means four twos multiplied: $2 \times 2 \times 2 \times 2 = 16$, then $16 + 1 = 17$.

B4

Question

$(5 + 1)^2 =$
Checked answer: 36

What the question is asking

This question asks for the value of $(5 + 1)^2$. Add inside the curved marks first, then multiply that answer by itself.

Important words

- *Parentheses* come first in PEMDAS.
- An *exponent* of 2 means square: multiply the number by itself.
- So 6^2 means 6×6.

Work the problem one tiny step at a time

1. Do P first: inside parentheses, $5 + 1 = 6$.
2. Now the expression is 6^2.
3. Do E next: $6^2 = 6 \times 6$.
4. Multiply: $6 \times 6 = 36$.

Final answer: 36

Why this makes sense

Parentheses first, then the square. Do not square 5 and 1 separately. $(5 + 1)^2$ is $6^2 = 36$, not $5^2 + 1^2 = 26$.

B5

Question

$3(2^2) =$

Checked answer: 12

What the question is asking

This question asks for the value of $3(2^2)$. A number next to curved marks means multiply.

Important words

- $3(2^2)$ means 3 times the value of (2^2).
- The *exponent* 2 in 2^2 means 2×2.
- PEMDAS: parentheses first, then exponents inside, then multiply.

Work the problem one tiny step at a time

1. Look inside the parentheses: 2^2.
2. Compute the exponent: $2^2 = 2 \times 2 = 4$.
3. Now the expression is $3(4)$, which means 3×4.
4. Multiply: $3 \times 4 = 12$.

Final answer: 12

Why this makes sense

The power is inside the parentheses, so square first, then multiply by 3. Notice this is different from $(3 \times 2)^2$, which would square after multiplying.

B6

Question

$(3 \times 2)^2 =$

Checked answer: 36

What the question is asking

This question asks for the value of $(3 \times 2)^2$. Multiply inside the curved marks first, then multiply that answer by itself.

Important words

- *Parentheses* first: multiply inside.
- Then the *exponent* 2 squares that result.

Work the problem one tiny step at a time

1. Inside parentheses: $3 \times 2 = 6$.
2. Now the expression is 6^2.
3. Square: $6^2 = 6 \times 6 = 36$.
4. Compare with B5: $3(2^2) = 12$, but $(3 \times 2)^2 = 36$. The parentheses change which step happens first.

Final answer: 36

Why this makes sense

Multiply inside first, then square. A common mix-up is treating this like $3(2^2)$. Those are different problems with different answers.

C. Decimals, Fractions, And Percents

C1

Question

In 82.731, the tenths digit is ____________
Checked answer: 7

What the question is asking

This question asks which single digit 0–9 sits in the first place after the dot in 82.731.

Important words

- A *digit* is a single symbol 0 through 9.
- A *decimal point* is the dot that separates whole numbers from parts less than one.
- *Tenths* means the first place to the right of the decimal point (one out of ten equal parts).
- *Hundredths* is the second place to the right. *Thousandths* is the third place to the right.

Work the problem one tiny step at a time

1. Write the number and name each place: 82.731.
2. Left of the point: 8 is tens, 2 is ones.
3. Right of the point, first digit is tenths: that digit is 7.
4. Next digit 3 is hundredths. Next digit 1 is thousandths.
5. The question asks only for tenths, so the answer is 7.

Final answer: 7

Why this makes sense

Tenths is always the first digit after the decimal point. Do not confuse tenths (7) with hundredths (3).

C2

Question

$314.7 + 29.58 =$ (decimal)
Checked answer: 344.28

What the question is asking

This question asks you to add two numbers that use a dot for parts of a whole, and write the answer the same way.

Important words

- A *decimal* is a number that uses a decimal point to show tenths, hundredths, and so on.
- The *sum* is the answer to an addition problem.
- To *add decimals*, line up the decimal points so equal place values sit in the same column.
- To *carry* means: when a column adds to 10 or more, write the ones digit in that column and move the extra tens digit to the next column on the left.

Work the problem one tiny step at a time

1. Write the numbers stacked, lining up the decimal points.
2. Write 314.7 as 314.70 so both numbers have two digits after the point.
3. Add hundredths: $0 + 8 = 8$.
4. Add tenths: $7 + 5 = 12$. Write 2 in tenths and carry 1 to the ones.
5. Add ones: $4 + 9 + 1$ (carry) $= 14$. Write 4, carry 1.
6. Add tens: $1 + 2 + 1$ (carry) $= 4$.
7. Add hundreds: 3.
8. Put the decimal point in the sum directly under the other points. The sum is 344.28:

$$\begin{array}{r} 314.70 \\ +\ 29.58 \\ \hline 344.28 \end{array}$$

Final answer: 344.28

Why this makes sense

Lining up decimal points keeps tenths with tenths and hundredths with hundredths. Check: $314.70 + 29.58$ should be a little more than $314 + 30 = 344$, and 344.28 fits.

C3

Question

$81.5 - 7.94 =$ (decimal)
Checked answer: 73.56

What the question is asking

This question asks you to subtract numbers that use a dot for parts of a whole, and write the answer the same way.

Important words

- The *difference* is the answer to a subtraction problem.
- To *subtract decimals*, line up the decimal points.
- You may write extra zeros so both numbers have the same number of decimal places.
- To *borrow* (also called *regrouping*) means: when a top digit is too small to subtract, take 1 from the next higher place on the left. That higher place decreases by 1, and the current place gains 10.

Work the problem one tiny step at a time

1. Write 81.5 as 81.50 to match hundredths.
2. Line up place values: 81.50 minus 7.94. Start digits are tens 8, ones 1, tenths 5, hundredths 0.
3. Hundredths: top digit 0 is smaller than 4, so borrow from tenths. Rewrite: tenths $5 \to 4$ (decrease by 1), hundredths $0 \to 10$ (gain 10). Now $10 - 4 = 6$.
4. Tenths: after the first borrow we have 4, and 4 is smaller than 9, so borrow from ones. Rewrite: ones $1 \to 0$, tenths $4 \to 14$. Now $14 - 9 = 5$.
5. Ones: after the second borrow we have 0, and 0 is smaller than 7, so borrow from tens. Rewrite: tens $8 \to 7$, ones $0 \to 10$. Now $10 - 7 = 3$.
6. Tens: $7 - 0 = 7$ (no more borrow needed).
7. Read the rewritten top digits and the difference: 73.56.

	tens	ones	.	tenths	hundredths
start top	8	1	.	5	0
after borrows	7	10	.	14	10
minus	0	7	.	9	4
difference	7	3	.	5	6

Each blue rewrite shows a borrow: higher place -1, current place $+10$.

Final answer: 73.56

Why this makes sense

Check by adding: $73.56 + 7.94 = 81.50$. That matches the starting number, so the subtraction is correct.

C4

Question
$0.32 =$ _________ %
Checked answer: 32%

What the question is asking
This question asks you to change 0.32 into an "out of 100" amount (written with %).

Important words

- A *percent* means "out of 100." The symbol is %.
- To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right) and write %.

Work the problem one tiny step at a time

1. Start with 0.32.
2. Multiply by 100: 0.32×100.
3. Move the point two places right: $0.32 \to 32$.
4. Add the percent symbol: 32%.

Final answer: 32%

Why this makes sense
Percent means hundredths. The decimal 0.32 is already 32 hundredths, so it is 32%.

C5

Question
$65\% =$ _________ as a decimal
Checked answer: 0.65

What the question is asking
This question asks you to change 65% into a number that uses a dot for parts of a whole.

Important words

- A *percent* is a number out of 100.
- A *fraction* $\frac{a}{b}$ means a equal parts out of b equal parts.
- To change a percent to a decimal, divide by 100 (move the decimal point two places to the left).

Work the problem one tiny step at a time

1. Start with 65%. Think of it as 65.0.
2. Divide by 100: $65 \div 100$.

3. Move the point two places left: $65.0 \to 0.65$.
4. So $65\% = 0.65$.

Final answer: 0.65

Why this makes sense

65% means 65 out of 100, which is the fraction $\frac{65}{100}$, and that decimal is 0.65.

C6

Question

$0.8 =$ __________ as a fraction

Checked answer: $\frac{4}{5}$

What the question is asking

This question asks you to write 0.8 as equal parts over a whole, then divide the top and bottom by the same common factor until nothing bigger than 1 divides both evenly.

Important words

- A *fraction* $\frac{a}{b}$ means a equal parts out of b equal parts.
- The *numerator* is the top number. The *denominator* is the bottom number.
- A *factor* of a number divides that number evenly (no leftover).
- A *common factor* divides both the top and the bottom.
- To *simplify* (simplifying) a fraction means divide the top and bottom by the same common factor to make an equal smaller fraction.
- *Simplest terms* means the top and bottom share no common factor greater than 1.
- The digit in the tenths place means "over 10."

Work the problem one tiny step at a time

1. The decimal 0.8 has 8 in the tenths place, so $0.8 = \frac{8}{10}$.
2. Find a common factor of 8 and 10. Both divide by 2.
3. Divide top and bottom by 2: $\frac{8 \div 2}{10 \div 2} = \frac{4}{5}$.
4. Check: 4 and 5 share no common factor greater than 1, so $\frac{4}{5}$ is simplest.

Final answer: $\frac{4}{5}$

Why this makes sense

0.8 is eight tenths. Simplifying $\frac{8}{10}$ by dividing by 2 gives $\frac{4}{5}$. Check: $4 \div 5 = 0.8$.

C7

Question
15% of \$80 =
Checked answer: \$12

What the question is asking
This question asks for 15% of \$80, meaning 15 hundredths of \$80.

Important words

- The word *of* in percent problems means multiply.
- 15% as a decimal is 0.15 (divide by 100).

Work the problem one tiny step at a time

1. Change 15% to a decimal: $15 \div 100 = 0.15$.
2. Multiply: $0.15 \times \$80$.
3. First compute 15×80. Break it: $15 \times 8 = 120$, then $15 \times 80 = 1200$ (one extra zero because $80 = 8 \times 10$).
4. There are two decimal places in 0.15, so place the point two places from the right in 1200: 12.00.
5. So $0.15 \times \$80 = \$12$.
6. Another way: 10% of \$80 is \$8, and 5% of \$80 is \$4, so 15% is $\$8 + \$4 = \$12$.

Final answer: \$12

Why this makes sense
Fifteen percent of \$80 is a little more than one tenth of \$80. One tenth of \$80 is \$8, so \$12 is a sensible size.

D. Facts And Decimal Multiply/Divide

D1

Question
$9 \times 6 =$
Checked answer: 54

What the question is asking
This question asks for the answer when you multiply 9 and 6.

Important words

- *Multiply* means find a total of equal groups.
- The *product* is the answer to a multiplication problem.

- A *factor* is a number being multiplied. Here the factors are 9 and 6.
- Equal groups: 9 groups of 6 means $6+6+6+6+6+6+6+6+6$.

Work the problem one tiny step at a time

1. Think of 9 groups of 6, or 6 groups of 9.
2. Use a known fact: $9 \times 5 = 45$.
3. Add one more group of 9: $45 + 9 = 54$.
4. Another path: $10 \times 6 = 60$, then take away one 6: $60 - 6 = 54$.
5. Check by dividing: $54 \div 9 = 6$ (because $9 \times 6 = 54$).

Final answer: 54

Why this makes sense
Check with addition: $6+6+6+6+6+6+6+6+6 = 54$. Or check $54 \div 9 = 6$.

D2

Question
$63 \div 9 =$
Checked answer: 7

What the question is asking
This question asks how many times 9 fits into 63.

Important words

- *Divide* means split into equal groups.
- The *quotient* is the answer to a division problem.
- The *dividend* is the number being divided (63).
- The *divisor* is the number you divide by (9).
- Division undoes multiplication: if $9 \times 7 = 63$, then $63 \div 9 = 7$.

Work the problem one tiny step at a time

1. Ask: what times 9 equals 63?
2. Try 9×6: $9 \times 6 = 54$. That is less than 63, so try one more.
3. Try 9×7: $9 \times 7 = 63$. Exact match.
4. So $63 \div 9 = 7$.
5. Check: multiply back — $9 \times 7 = 63$.

Final answer: 7

Why this makes sense

Multiplication check: $9 \times 7 = 63$. The quotient must be 7.

D3

Question

$2.4 \times 1.5 =$___________ (decimal)
Checked answer: 3.6

What the question is asking

This question asks you to multiply two numbers that use a dot for parts of a whole.

Important words

- A *factor* is a number being multiplied. Here the factors are 2.4 and 1.5.
- A *decimal digit* is a digit to the right of the decimal point (after the dot).
- To multiply decimals: multiply as if they were whole numbers, then place the decimal point by counting all decimal digits in the factors.

Work the problem one tiny step at a time

1. Ignore the points for a moment: multiply 24×15.
2. $24 \times 10 = 240$.
3. $24 \times 5 = 120$.
4. Add: $240 + 120 = 360$.
5. Count decimal digits: 2.4 has 1 digit after the point; 1.5 has 1 digit after the point; total 2 digits.
6. Place the point two places from the right in 360: 3.60.
7. You may drop the trailing zero: $3.60 = 3.6$.

Final answer: 3.6

Why this makes sense

Estimate: $2 \times 1.5 = 3$, and 2.4 is a bit more than 2, so about 3.6 makes sense.

D4

Question

$5.6 \div 0.8 =$___________ (decimal)
Checked answer: 7

What the question is asking

This question asks you to divide 5.6 by 0.8.

Important words

- The *dividend* is the number being divided (5.6).
- The *divisor* is the number you divide by (0.8).
- The *quotient* is the answer to a division problem.
- A *power of ten* is a number like 10, 100, or 1000 made by multiplying tens together.
- To divide by a decimal, multiply the dividend and the divisor by the same power of ten so the divisor becomes a whole number.

Work the problem one tiny step at a time

1. Name the parts: dividend 5.6, divisor 0.8.
2. Multiply both 5.6 and 0.8 by 10 to clear one decimal place in the divisor.
3. $5.6 \times 10 = 56$ and $0.8 \times 10 = 8$.
4. Now compute $56 \div 8$.
5. $8 \times 7 = 56$, so $56 \div 8 = 7$.
6. Therefore $5.6 \div 0.8 = 7$.

Final answer: 7

Why this makes sense

Check: $0.8 \times 7 = 5.6$. Yes, the quotient 7 is correct.

E. Ratios, Fractions, And Proportions

E1

Question

Greater: $\frac{5}{8}$ or 0.61?

Checked answer: $\frac{5}{8}$

What the question is asking

This question asks which number is larger: $\frac{5}{8}$ or 0.61.

Important words

- To *compare* a fraction and a decimal, write both in the same form (both decimals or both fractions).
- A *fraction bar* means divide: $\frac{5}{8}$ means $5 \div 8$.
- A *remainder* is what is left over after a division step (how much is still not divided).

- To *bring down* a digit means move the next digit (often a 0 after the decimal) into the current remainder so you can keep dividing.

Work the problem one tiny step at a time

1. Change $\frac{5}{8}$ to a decimal by dividing $5 \div 8$.
2. 8 into 50 is 6 (because $8 \times 6 = 48$), remainder 2.
3. Bring down 0: $20 \div 8 = 2$ (because $8 \times 2 = 16$), remainder 4.
4. Bring down 0: $40 \div 8 = 5$. So $5 \div 8 = 0.625$.
5. Compare 0.625 and 0.61. Line up: 0.625 versus 0.610.
6. Tenths: both have 6. Hundredths: $2 > 1$, so $0.625 > 0.610$ already (thousandths $5 > 0$ also agrees).
7. Therefore $\frac{5}{8}$ is greater.

Final answer: $\frac{5}{8}$

Why this makes sense

0.625 is a little more than 0.61 because the hundredths digit 2 beats 1. So the fraction $\frac{5}{8}$ wins.

E2

Question

$$\frac{4}{7} = \frac{?}{21} \qquad ? =$$

Checked answer: 12

What the question is asking

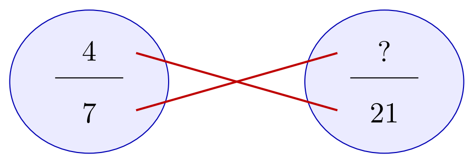
This question asks for the missing top number that makes the two equal-parts-over-a-whole amounts equal.

Important words

- *Equal fractions* name the same amount, even if the numbers look different.
- A *ratio* compares two amounts. You can write a ratio as a fraction, like $\frac{4}{7}$.
- An *equation* is a math sentence with an equals sign that says two sides are equal.
- A *proportion* is an equation that says two ratios (fractions) are equal.
- The *butterfly method* (cross multiply) multiplies along the diagonals of two equal fractions.
- *Cross products* are the two diagonal products from the butterfly ($a \times d$ and $b \times c$).
- The *numerator* is the top. The *denominator* is the bottom.

Work the problem one tiny step at a time

1. Write the equal fractions: $\frac{4}{7} = \frac{?}{21}$.
2. Draw the butterfly (cross-multiply picture) below.
3. Cross multiply along the red lines: $4 \times 21 = 7 \times ?$.
4. Compute 4×21: $4 \times 20 = 80$ and $4 \times 1 = 4$, so 84.
5. So $84 = 7 \times ?$.
6. Divide both sides by 7 (shown under the equals):



Butterfly: multiply along each red line.

$$84 = 7 \times ?$$
$$\underline{\div 7 \qquad \div 7}$$

7. Compute the leftover division: $84 \div 7 = 12$ (because $7 \times 12 = 84$).
8. So $? = 12$.

Final answer: 12

Why this makes sense

Check another way: $7 \times 3 = 21$, so multiply the top by the same 3: $4 \times 3 = 12$. Same answer. Cross products match: $4 \times 21 = 84$ and $7 \times 12 = 84$.

E3

Question

4 snacks cost \$20. 7 snacks cost ____________ dollars.
Checked answer: 35 dollars

What the question is asking

This question asks how many dollars 7 snacks cost if 4 snacks cost \$20 and the price stays in the same pattern.

Important words

- A *ratio* compares two amounts (here, snacks to dollars).
- A *proportion* keeps two ratios equal.
- A *unit price* is the cost of one item.
- We use the butterfly (cross multiply) when two fractions are equal.

Work the problem one tiny step at a time

1. Set up a proportion. Keep snacks on top and dollars on bottom (or keep the same order on both sides).

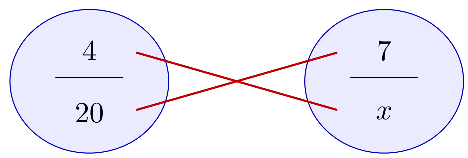
2. One correct setup:
$$\frac{4 \text{ snacks}}{20 \text{ dollars}} = \frac{7 \text{ snacks}}{x \text{ dollars}}$$

3. Draw the butterfly and cross multiply: $4 \times x = 20 \times 7$.

4. Break 20×7: $20 \times 5 = 100$ and $20 \times 2 = 40$, then $100 + 40 = 140$.

5. So $4x = 140$.

6. Divide both sides by 4:



Butterfly: multiply along each red line.

$$4x = 140$$
$$\underline{\div 4 \qquad \div 4}$$

7. Compute the leftover division: $140 \div 4 = 35$ (because $4 \times 35 = 140$).

8. So $x = 35$.

Final answer: 35 dollars

Why this makes sense

Unit-price check: one snack costs $20 \div 4 = 5$ dollars. Then 7 snacks cost $7 \times 5 = 35$ dollars. Same answer.

E4

Question

$$\frac{5}{6} = \frac{x}{18} \qquad x =$$

Checked answer: $x = 15$

What the question is asking

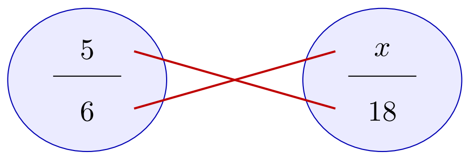
This question asks for the value of x that makes the two equal-parts-over-a-whole amounts equal.

Important words

- x is a *variable*: a letter that stands for an unknown number.
- Use the *butterfly method* (cross multiply) for equal fractions.

Work the problem one tiny step at a time

1. Write $\dfrac{5}{6} = \dfrac{x}{18}$.
2. Cross multiply: $5 \times 18 = 6 \times x$.
3. Compute $5 \times 18 = 90$.
4. So $90 = 6x$.
5. Divide both sides by 6:



Butterfly: multiply along each red line.

$$90 = 6x$$

$$\underline{\div 6 \qquad \div 6}$$

6. Compute the leftover division: $90 \div 6 = 15$ (because $6 \times 15 = 90$).
7. So $x = 15$.

Final answer: $x = 15$

Why this makes sense

Check: $6 \times 3 = 18$, so multiply the top by 3: $5 \times 3 = 15$. Cross products: $5 \times 18 = 90$ and $6 \times 15 = 90$.

F. Vocabulary, Expressions, And Equations

F1

Question

In $7m$, 7 is the ________________
Checked answer: coefficient

What the question is asking

This question asks for the name of the number 7 in $7m$ (a times pairing of a number and a letter).

Important words

- A *variable* is a letter that stands for a number. Here m is a variable.
- A *coefficient* is the number that multiplies a variable. In $7m$, the 7 multiplies m.
- A *term* is a product of numbers and variables, like $7m$.

Work the problem one tiny step at a time

1. Look at $7m$. This means $7 \times m$.
2. The letter m is the variable.
3. The number in front that multiplies the variable is 7.
4. That number is called the coefficient.

Final answer: coefficient

Why this makes sense
In $7m$, think "7 copies of m." The 7 is the coefficient.

F2

Question
In $n + 8 = 12$, n is the ________________
Checked answer: variable

What the question is asking
This question asks what we call the letter n in the math sentence $n + 8 = 12$ (two sides joined by an equals sign).

Important words

- An *equation* is a math sentence with an equals sign that says two sides are equal.
- A *variable* is a letter that stands for an unknown number we can find.
- A *constant* (here) is a fixed known number, like 8 or 12 in this equation.
- The letter n is not a digit you can read off the page; it stands for a value you still have to find.

Work the problem one tiny step at a time

1. Look at the equation $n + 8 = 12$.
2. The numbers 8 and 12 are known constants.
3. The letter n stands for some unknown number.
4. We do not yet know which number n is, but the equation tells us how to find it.
5. A letter used this way is called a variable.
6. Numeric check of the idea: if $n = 4$, then $4 + 8 = 12$, which matches the equation. The letter was standing for that unknown.

Final answer: variable

Why this makes sense
n is not a fixed known digit like 8 or 12. It is the unknown, so it is a variable. Check: $4 + 8 = 12$ shows the letter can be replaced by a number that makes the equation true.

F3

Question
Undo $+8$, use inverse op: ____________
Checked answer: subtract 8 (or -8)

What the question is asking
This question asks which operation undoes adding 8.

Important words

- An *inverse operation* undoes another operation (takes you back to the start).
- *Addition* means plus (written $+$). *Subtraction* means take away (written $-$).
- Addition and subtraction are inverse operations of each other.
- To *cancel* (here) means two inverse moves undo each other and leave the starting value.

Work the problem one tiny step at a time

1. Starting operation: add 8 (written $+8$).
2. The inverse of add is subtract.
3. To undo adding 8, subtract 8 (written -8).
4. Numeric round-trip: start with 10, add 8 to get 18, then subtract 8 to get back to 10.

Final answer: subtract 8 (or -8)

Why this makes sense
Plus and minus cancel (they undo each other). The inverse of $+8$ is -8.

F4

Question
Undo $\times 5$, use inverse op: ____________
Checked answer: divide by 5 (or $\div 5$)

What the question is asking
This question asks which operation undoes multiplying by 5.

Important words

- An *inverse operation* undoes another operation (takes you back to the start).
- *Multiplication* means times (written $\times$). *Division* means split into equal groups (written $\div$).
- Multiplication and division are inverse operations of each other.
- To *cancel* (here) means two inverse moves undo each other and leave the starting value (same idea as F3).

Work the problem one tiny step at a time

1. Starting operation: multiply by 5 (written $\times 5$).
2. The inverse of multiply is divide.
3. To undo multiplying by 5, divide by 5 (written $\div 5$).
4. Numeric round-trip: start with 4, multiply by 5 to get 20, then divide by 5 to get back to 4.

Final answer: divide by 5 (or $\div 5$)

Why this makes sense
Multiply and divide cancel (they undo each other, same idea as F3). The inverse of $\times 5$ is $\div 5$.

F5

Question
Combine like terms: $6y - 2y =$
Checked answer: $4y$

What the question is asking
This question asks you to rewrite $6y - 2y$ in a shorter form by adding or subtracting the numbers in front of matching y parts.

Important words

- To *simplify* an expression means rewrite it in a shorter equivalent form (here, one term instead of two).
- *Like terms* are terms that have the exact same variable part (here both have y).
- To *combine like terms*, add or subtract the coefficients and keep the same variable.

Work the problem one tiny step at a time

1. Both terms have the variable y, so they are like terms.
2. Subtract the coefficients: $6 - 2 = 4$.
3. Keep the variable y: the result is $4y$.
4. Think: 6 copies of y take away 2 copies of y leaves 4 copies of y.

Final answer: $4y$

Why this makes sense
You only combine the numbers in front. You do not invent y^2 or drop the y. The answer stays $4y$.

F6

Question
If $n = 4$, then $5n - 3 =$
Checked answer: 17

What the question is asking
This question asks you to find the value of $5n - 3$ when n is 4 (put 4 in for n and compute).

Important words

- To *evaluate* means replace the variable with a number and compute.
- *PEMDAS*: multiply before you subtract.
- $5n$ means $5 \times n$.

Work the problem one tiny step at a time

1. Replace n with 4: write $5(4) - 3$.
2. Multiply first: $5 \times 4 = 20$.
3. Now the expression is $20 - 3$.
4. Subtract: $20 - 3 = 17$.

Final answer: 17

Why this makes sense
Do not subtract $4 - 3$ first. Multiply 5×4 first because multiplication comes before subtraction.

G. Distributive Property

G1

Question
Expand: $3(y + 6) =$
Checked answer: $3y + 18$

What the question is asking
This question asks you to rewrite $3(y + 6)$ without curved marks by multiplying the outside number by each inside part.

Important words

- The *distributive property* says $a(b + c) = ab + ac$. You multiply the outside number by each part inside.
- To *expand* means write the expression without the parentheses by distributing.

Work the problem one tiny step at a time

1. The outside number is 3. The inside parts are y and 6.
2. Multiply 3 by the first inside part: $3 \times y = 3y$.
3. Multiply 3 by the second inside part: $3 \times 6 = 18$.
4. Add those products: $3y + 18$.

Final answer: $3y + 18$

Why this makes sense
Check with a number: if $y = 2$, left side $3(2 + 6) = 3(8) = 24$. Right side $3(2) + 18 = 6 + 18 = 24$. Same value.

H. Equations, Inequalities, And Graphing

H1

Question
$x + 7 = 18 \qquad x =$
Checked answer: $x = 11$

What the question is asking
This question asks you to find the number x that makes both sides equal.

Important words

- An *equation* says two sides are equal.
- To *solve* means find the value of the variable.
- Use the *inverse operation*: undo $+7$ by subtracting 7 from both sides.
- To *substitute* means put the number back in for the letter and check both sides.

Work the problem one tiny step at a time

1. Goal: get x alone on one side so both sides stay equal.
2. Start with $x + 7 = 18$.
3. The left side shows x plus 7. The inverse of add 7 is subtract 7.
4. Subtract 7 from both sides. Show the operation under the equals:

$$\begin{array}{c} x + 7 = 18 \\ \underline{\;-7 \qquad -7\;} \end{array}$$

5. Left side: $x + 7 - 7$ leaves x.
6. Right side leftover arithmetic: $18 - 7 = 11$.

7. So $x = 11$.

8. Check by substituting: put 11 back in for x. Compute $11 + 7 = 18$. Both sides match.

Final answer: $x = 11$

Why this makes sense

Check: put 11 back in. $11 + 7 = 18$. Both sides match, so $x = 11$ is correct.

H2

Question

$\frac{x}{4} = 6 \qquad x =$

Checked answer: $x = 24$

What the question is asking

This question asks you to find the number x that makes $\frac{x}{4} = 6$ true.

Important words

- $\frac{x}{4}$ means x divided by 4.
- The inverse of divide by 4 is multiply by 4. Do that to both sides.

Work the problem one tiny step at a time

1. Goal: get x alone on one side so both sides stay equal.

2. Start with $\frac{x}{4} = 6$.

3. The left side shows x divided by 4. The inverse of divide by 4 is multiply by 4.

4. Multiply both sides by 4. Show the operation under the equals:

$$\begin{array}{ccc} \frac{x}{4} & = & 6 \\ \underline{\times 4} & & \underline{\times 4} \end{array}$$

5. Left side: $\frac{x}{4} \times 4$ leaves x.

6. Right side leftover arithmetic: $6 \times 4 = 24$.

7. So $x = 24$.

8. Check by substituting: put 24 back in for x. Compute $\frac{24}{4} = 6$. Both sides match.

Final answer: $x = 24$

Why this makes sense

Check: $\frac{24}{4} = 6$. Yes, both sides match.

H3

Question

Solve: $x + 3 > 8$
Checked answer: $x > 5$

What the question is asking

This question asks you to find which numbers for x make the left side of $x + 3 > 8$ greater than the right.

Important words

- An *inequality* compares two sides with a symbol like $>$ (greater than) or $<$ (less than).
- $>$ means "greater than" (left side is bigger).
- To *isolate* x means get the letter x alone on one side.
- You can add or subtract the same number on both sides of an inequality, just like an equation.

Work the problem one tiny step at a time

1. Start with $x + 3 > 8$.
2. Subtract 3 from both sides to isolate x. Show under the inequality:

$$\begin{array}{ccc} x+3 & > & 8 \\ \underline{-3} & & \underline{-3} \end{array}$$

3. Left side: $x + 3 - 3$ leaves x.
4. Right side leftover arithmetic: $8 - 3 = 5$.
5. So $x > 5$.

Final answer: $x > 5$

Why this makes sense

Any number greater than 5 works. Example: if $x = 6$, then $6 + 3 = 9$ and $9 > 8$ is true. If $x = 5$, then $5 + 3 = 8$, and $8 > 8$ is false, so 5 is not included.

H4

Question

Solve: $x - 4 < 2$
Checked answer: $x < 6$

What the question is asking

This question asks you to find which numbers for x make the left side of $x - 4 < 2$ smaller than the right.

Important words

- $<$ means "less than" (left side is smaller).
- Undo -4 by adding 4 to both sides.

Work the problem one tiny step at a time

1. Start with $x - 4 < 2$.
2. Add 4 to both sides. Show under the inequality:

$$\begin{array}{c} x - 4 < 2 \\ \underline{\;+4 \quad\; +4\;} \end{array}$$

3. Left side: $x - 4 + 4$ leaves x.
4. Right side leftover arithmetic: $2 + 4 = 6$.
5. So $x < 6$.

Final answer: $x < 6$

Why this makes sense

Example check: if $x = 5$, then $5 - 4 = 1$ and $1 < 2$ is true. If $x = 6$, then $6 - 4 = 2$ and $2 < 2$ is false, so 6 is not included.

H5

Question

Write an equivalent way to express $6 < y$:
Checked answer: $y > 6$

What the question is asking

This question asks for another comparison sentence that means the exact same thing as $6 < y$.

Important words

- *Equivalent* means "means the same thing."
- To *flip* an inequality means swap the two sides and reverse the symbol ($<$ becomes $>$, or $>$ becomes $<$).
- If you swap the two sides of an inequality, you must also flip the inequality symbol.
- $6 < y$ means 6 is less than y, which is the same as saying y is greater than 6.

Work the problem one tiny step at a time

1. Read $6 < y$: "six is less than y."
2. That means y is to the right of 6 on the number line.
3. Flip both sides and flip $<$ to $>$: $y > 6$.

4. Read $y > 6$: "y is greater than six." Same meaning.

Final answer: $y > 6$

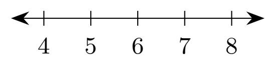
Why this makes sense

Do not write $y < 6$. That would reverse the meaning. Flipping sides requires flipping the symbol.

H6

Question

Graph $6 < y$ on a number line:



Checked answer: Open circle at 6, shade to the right (same as $y > 6$).

What the question is asking

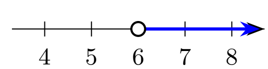
This question asks you to show all numbers y that are greater than 6 on a straight line of numbers.

Important words

- An *endpoint* is the starting number on the graph where the circle sits.
- An *open circle* means the endpoint is *not* included (used for $<$ or $>$).
- A *closed* (filled) circle would mean the endpoint is included (used for $\leq$ or $\geq$).
- A *ray* on a number line is a shaded half-line that starts at a point and keeps going forever in one direction (shown with an arrow).
- Shading to the *right* means greater numbers. Shading to the *left* means smaller numbers.
- $6 < y$ means the same as $y > 6$.

Work the problem one tiny step at a time

1. Rewrite as $y > 6$ so y is on the left (easier to graph).
2. Find 6 on the number line.
3. Draw an open circle at 6 because y cannot equal 6.
4. Shade the ray to the right of 6, because numbers greater than 6 are to the right.
5. Put an arrow on the shaded part to show it keeps going.



Final answer: Open circle at 6, shade right.

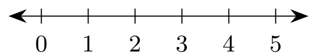
Why this makes sense

Open (not filled) circle matters. If the circle were filled, that would wrongly include 6.

H7

Question

Graph $x > 2$ on a number line:



Checked answer: Open circle at 2, shade to the right.

What the question is asking

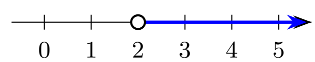
This question asks you to show all numbers x greater than 2 on a straight line of numbers.

Important words

- $x > 2$ means x is greater than 2 (not equal to 2).
- Use an *open circle* at 2 and shade to the *right*.

Work the problem one tiny step at a time

1. Find 2 on the number line.
2. Draw an open circle at 2 (because $>$ does not include 2).
3. Shade to the right of 2 for all larger numbers.
4. Add an arrow to show the shading continues forever.



Final answer: Open circle at 2, shade right.

Why this makes sense

Numbers like 3, 4, 10, and 100 are shaded. The number 2 itself is not shaded.

I. Geometry

I1

Question

Rectangle with length 9 and width 4: Area =
Checked answer: 36 square units

What the question is asking

This question asks how much flat space is inside a rectangle with length 9 and width 4.

Important words

- A *rectangle* is a four-sided shape with four right angles. Opposite sides are equal.
- *Length* and *width* are the two side lengths.

- *Area* means how much flat space is inside. For a rectangle, area = length $\times$ width.
- Area is measured in *square units*.

Work the problem one tiny step at a time

1. Write the formula: area = length $\times$ width.
2. Put in the numbers: area = 9×4.
3. Multiply: $9 \times 4 = 36$.
4. Include the unit idea: 36 square units.

Final answer: 36 square units

Why this makes sense
Area counts square tiles that fit inside. A 9-by-4 grid holds 36 unit squares.

I2

Question
Rectangle with length 11 and width 3: Perimeter =
Checked answer: 28 units

What the question is asking
This question asks for the distance all the way around a rectangle with length 11 and width 3.

Important words

- *Perimeter* means the distance all the way around the outside.
- For a rectangle, perimeter = 2(length + width), or add all four sides.

Work the problem one tiny step at a time

1. Add length and width: $11 + 3 = 14$.
2. There are two of each side, so multiply by 2: $2 \times 14 = 28$.
3. Or add all four sides: $11 + 3 + 11 + 3 = 28$.

Final answer: 28 units

Why this makes sense
Perimeter is a length around the edge, not the space inside. Do not multiply 11×3 here; that would be area.

I3

Question

Rectangle with area 42 sq units and width 6: Length =
Checked answer: 7 units

What the question is asking

This question asks for the missing length when the flat space inside and the width are known.

Important words

- Area = length $\times$ width.
- So length = area $\div$ width.

Work the problem one tiny step at a time

1. Write the area equation: $42 = \text{length} \times 6$.
2. Divide both sides by 6 to find length. Show the operation under the equals:

$$\begin{array}{ccc} 42 & = & \text{length} \times 6 \\ \hline \div 6 & & \div 6 \end{array}$$

3. Compute the leftover division: $42 \div 6 = 7$ (because $6 \times 7 = 42$).
4. So length = 7 units.

Final answer: 7 units

Why this makes sense

Check: $7 \times 6 = 42$. The area matches.

J. Units And Quantities

J1

Question

3 hours = ____________ minutes
Checked answer: 180 minutes

What the question is asking

This question asks you to change 3 hours into minutes.

Important words

- A *conversion* changes a measurement from one unit to another.
- The key fact: 1 hour = 60 minutes.

Work the problem one tiny step at a time

1. Start with 3 hours.
2. Each hour has 60 minutes, so multiply: 3×60.
3. $3 \times 60 = 180$.
4. So 3 hours = 180 minutes.

Final answer: 180 minutes

Why this makes sense

Three groups of 60 minutes make 180 minutes. Do not divide; hours to minutes gets a larger number of smaller units.

J2

Question

$\frac{3}{5}$ of 35 =

Checked answer: 21

What the question is asking

This question asks for $\frac{3}{5}$ of 35 (three fifths of 35).

Important words

- The word *of* with a fraction means multiply.
- $\frac{3}{5}$ means 3 equal parts out of 5 equal parts.

Work the problem one tiny step at a time

1. Write the product: $\frac{3}{5} \times 35$.
2. One helpful way: first find $\frac{1}{5}$ of 35 by dividing $35 \div 5 = 7$.
3. Then $\frac{3}{5}$ means 3 of those parts: $3 \times 7 = 21$.
4. Or: $\frac{3 \times 35}{5} = \frac{105}{5} = 21$.

Final answer: 21

Why this makes sense

One fifth of 35 is 7, so three fifths is 21. That is a bit more than half of 35 (half is 17.5), and 21 fits.

J3

Question

210 miles/3 hours =
Checked answer: 70 miles per hour

What the question is asking

This question asks how many miles go with 1 hour (the amount for one hour).

Important words

- A *unit rate* tells how much of one quantity goes with 1 of another quantity.
- *Miles per hour* means miles for each 1 hour.
- To find a unit rate, divide the first amount by the second amount.
- Here: miles $\div$ hours gives miles for each 1 hour.

Work the problem one tiny step at a time

1. You have 210 miles and 3 hours.
2. Divide miles by hours to get miles for 1 hour: $210 \div 3$.
3. $3 \times 70 = 210$, so $210 \div 3 = 70$.
4. The unit rate is 70 miles per hour.
5. Check: 70 miles per hour $\times$ 3 hours = 210 miles.

Final answer: 70 miles per hour

Why this makes sense

In 3 hours you go 210 miles, so in 1 hour you go one-third of that: 70 miles. Multiplying back $70 \times 3 = 210$ confirms the unit rate.

J4

Question

72 inches = ____________ yards
Checked answer: 2 yards

What the question is asking

This question asks you to change 72 inches into yards.

Important words

- A *unit conversion* changes a measurement from one unit name to another equal amount.
- An *inch* is a small length unit. A *yard* is a larger length unit.
- The key fact: 1 yard = 36 inches.

- To go from a smaller unit (inches) to a larger unit (yards), divide by how many small units fit in one large unit.

Work the problem one tiny step at a time

1. Write what you know: 72 inches, and 1 yard = 36 inches.
2. Divide inches by 36: $72 \div 36$.
3. $36 \times 2 = 72$, so $72 \div 36 = 2$.
4. So 72 inches = 2 yards.
5. Check: 2 yards $\times$ 36 inches per yard = 72 inches.

Final answer: 2 yards

Why this makes sense

Two yards means $36 + 36 = 72$ inches. Check passes: $2 \times 36 = 72$.

J5

Question

150 miles/5 gallons =
Checked answer: 30 miles per gallon

What the question is asking

This question asks how many miles go with 1 gallon.

Important words

- A *unit rate* tells how much of one quantity goes with 1 of another quantity.
- *Miles per gallon* means how many miles you travel on 1 gallon.
- To find a unit rate, divide the first amount by the second amount.
- Here: miles $\div$ gallons gives miles for each 1 gallon.

Work the problem one tiny step at a time

1. You have 150 miles and 5 gallons.
2. Divide miles by gallons to get miles for 1 gallon: $150 \div 5$.
3. $5 \times 30 = 150$, so $150 \div 5 = 30$.
4. The unit rate is 30 miles per gallon.
5. Check: 30 miles per gallon $\times$ 5 gallons = 150 miles.

Final answer: 30 miles per gallon

Why this makes sense

Five gallons take you 150 miles, so one gallon takes you 30 miles.

K. Expressions, Factors, And Fraction Structure

K1

Question

Is $8x + 1$ a linear expression?
Checked answer: Yes

What the question is asking

This question asks whether $8x + 1$ has the letter x only to the first power (a small raised 1; no x^2, and no letter under a top-over-bottom bar).

Important words

- An *expression* is a math phrase with numbers, variables, and operations (no equals sign required).
- A *linear expression* has variables only to the first power. There is no x^2, no x^3, and no variable in a denominator. Typical form: $ax + b$.
- *First power* means a small raised 1 on the variable (written x^1, or just x).
- A *constant* is a number that stands alone with no variable letter (like 1 or 8).
- The term $8x$ means $8 \times x$, and x is to the first power.

Work the problem one tiny step at a time

1. Look at each term in $8x + 1$.
2. The term $8x$ has variable x to the power 1.
3. The term 1 is a constant (no variable).
4. There is no x^2 or higher power.
5. So $8x + 1$ is a linear expression. Answer: Yes.

Final answer: Yes

Why this makes sense

Contrast with $8x^2 + 1$, which is not linear because of x^2. Here the power of x is 1, so it is linear.

K2

Question

Prime factorization of $90 =$
Checked answer: $2 \cdot 3^2 \cdot 5$

What the question is asking

This question asks you to write 90 as a times-chain of numbers that each can be divided evenly only by 1 and themselves.

Important words

- A *factor* of a number divides that number evenly (no leftover).
- A *prime number* has exactly two factors: 1 and itself. Examples: $2, 3, 5, 7, 11$.
- *Prime factorization* writes a number as a product of primes only.
- An *exponent* like 3^2 means 3×3.

Work the problem one tiny step at a time

1. Factor 90 into any pair: $90 = 9 \times 10$, or better, start dividing by small primes.
2. 90 is even, so divide by 2: $90 = 2 \times 45$.
3. $45 = 5 \times 9$, and $9 = 3 \times 3$.
4. So $90 = 2 \times 3 \times 3 \times 5$.
5. Write repeated factors with an exponent: $2 \cdot 3^2 \cdot 5$.

Final answer: $2 \cdot 3^2 \cdot 5$

Why this makes sense
Check by multiplying: $3^2 = 9$, then $2 \times 9 = 18$, then $18 \times 5 = 90$. Correct.

K3

Question
Find the GCF of 27 and 36:
Checked answer: 9

What the question is asking
This question asks for the biggest whole number that divides both 27 and 36 evenly (no leftover).

Important words

- A *factor* of a number divides that number evenly.
- A *common factor* divides both numbers.
- The *GCF* (greatest common factor) is the largest common factor.
- To *prime factorize* means write the number as a product of prime numbers only.
- *Shared primes* are prime factors that appear in both prime factorizations.

Work the problem one tiny step at a time

1. Prime factorize 27: $27 = 3 \times 3 \times 3 = 3^3$.
2. Prime factorize 36: $36 = 2 \times 2 \times 3 \times 3 = 2^2 \cdot 3^2$.
3. Shared primes: only 3 appears in both; the smaller power is 3^2.

4. So GCF $= 3^2 = 9$.
5. List check: factors of 27: $1, 3, 9, 27$. Factors of 36: $1, 2, 3, 4, 6, 9, 12, 18, 36$. Largest shared: 9.

Final answer: 9

Why this makes sense

9 divides 27 ($27 \div 9 = 3$) and divides 36 ($36 \div 9 = 4$). Nothing larger than 9 does both.

K4

Question

Simplify $\frac{27}{36} =$

Checked answer: $\frac{3}{4}$

What the question is asking

This question asks you to shrink $\frac{27}{36}$ by dividing the top and bottom by the same whole number until nothing bigger than 1 divides both evenly.

Important words

- To *simplify* a fraction, divide top and bottom by their GCF.
- *Simplest terms* means the top and bottom share no common factor greater than 1.
- From K3, the GCF of 27 and 36 is 9.

Work the problem one tiny step at a time

1. From K3, the GCF of 27 and 36 is 9 (shared primes give $3^2 = 9$).
2. Divide the numerator by 9: $27 \div 9 = 3$.
3. Divide the denominator by 9: $36 \div 9 = 4$.
4. Write $\frac{3}{4}$.
5. Check: 3 and 4 share no common factor greater than 1, so it is simplest.

Final answer: $\frac{3}{4}$

Why this makes sense

$\frac{27}{36}$ and $\frac{3}{4}$ name the same amount. Cross check: $27 \times 4 = 108$ and $36 \times 3 = 108$.

K5

Question

Convert $2\frac{2}{3}$ to an improper fraction:

Checked answer: $\frac{8}{3}$

What the question is asking

This question asks you to rewrite $2\frac{2}{3}$ (a whole number plus equal parts over a whole) as one top-over-bottom amount where the top is at least as big as the bottom.

Important words

- A *mixed number* has a whole number and a fraction, like $2\frac{2}{3}$.
- An *improper fraction* has a numerator greater than or equal to the denominator, like $\dfrac{8}{3}$.
- Rule: improper numerator = (whole × denominator) + numerator. Keep the same denominator.

Work the problem one tiny step at a time

1. Whole number = 2, numerator = 2, denominator = 3.
2. Multiply whole by denominator: $2 \times 3 = 6$.
3. Add the old numerator: $6 + 2 = 8$.
4. Keep denominator 3: $\dfrac{8}{3}$.

Final answer: $\dfrac{8}{3}$

Why this makes sense

Check: $\dfrac{8}{3} = 2 + \dfrac{2}{3}$ because $\dfrac{6}{3} = 2$ and $\dfrac{2}{3}$ remains. Same mixed number.

K6

Question

Which is greater: $1\frac{3}{4}$ or $(1)\left(\frac{3}{4}\right)$?

Checked answer: $1\frac{3}{4}$

What the question is asking

This question asks which value is larger: $1\frac{3}{4}$ (one and three fourths) or $(1)\left(\frac{3}{4}\right)$ (one times three fourths).

Important words

- $1\frac{3}{4}$ means $1 + \dfrac{3}{4}$.
- $(1)\left(\frac{3}{4}\right)$ means 1 times $\dfrac{3}{4}$, which is just $\dfrac{3}{4}$.
- Parentheses next to each other mean multiply.

Work the problem one tiny step at a time

1. Rewrite the mixed number: $1\frac{3}{4} = 1 + \dfrac{3}{4} = \dfrac{4}{4} + \dfrac{3}{4} = \dfrac{7}{4}$.

2. Compute the product: $(1)\left(\frac{3}{4}\right) = \dfrac{3}{4}$.

3. Compare $\dfrac{7}{4}$ and $\dfrac{3}{4}$. Same denominator, so compare tops: $7 > 3$.

4. Therefore $1\frac{3}{4}$ is greater.

Final answer: $1\frac{3}{4}$

Why this makes sense

A mixed number includes a whole 1 plus a fraction. Multiplying by 1 does not add a whole; it only keeps $\dfrac{3}{4}$. So the mixed number is larger.

K7

Question

In $\dfrac{m}{2}$, coeff of m is _________

Checked answer: $\dfrac{1}{2}$

What the question is asking

This question asks for the number that multiplies the letter m in $\dfrac{m}{2}$.

Important words

- A *coefficient* is the number multiplying a variable.
- $\dfrac{m}{2}$ means m divided by 2, which is the same as $\dfrac{1}{2} \times m$.

Work the problem one tiny step at a time

1. Rewrite $\dfrac{m}{2}$ as $\dfrac{1}{2}m$.

2. The number multiplying m is $\dfrac{1}{2}$.

3. So the coefficient of m is $\dfrac{1}{2}$.

Final answer: $\dfrac{1}{2}$

Why this makes sense

Dividing by 2 is the same as multiplying by one-half. The coefficient is not 2; it is $\dfrac{1}{2}$.

K8

Question
In $\frac{q}{5}$, coeff of q is __________
Checked answer: $\frac{1}{5}$

What the question is asking

This question asks for the number that multiplies the letter q in $\frac{q}{5}$.

Important words

- A *variable* is a letter that stands for a number. Here the variable is q.
- A *coefficient* is the number that multiplies a variable.
- A *fraction* $\frac{q}{5}$ means q divided by 5.
- Dividing by 5 is the same as multiplying by $\frac{1}{5}$: $\frac{q}{5} = \frac{1}{5} \times q$.

Work the problem one tiny step at a time

1. Start with $\frac{q}{5}$.
2. Rewrite dividing by 5 as multiplying by $\frac{1}{5}$: $\frac{q}{5} = \frac{1}{5}q$.
3. The number multiplying q is $\frac{1}{5}$.
4. So the coefficient is $\frac{1}{5}$.
5. Numeric check: if $q = 10$, then $\frac{10}{5} = 2$ and $\frac{1}{5} \times 10 = 2$. Same value, so $\frac{1}{5}$ really is the multiplier.

Final answer: $\frac{1}{5}$

Why this makes sense

Dividing by 5 means multiplying by $\frac{1}{5}$. The coefficient is that multiplier. Check with $q = 10$: both forms give 2.

K9

Question
Commutative: $8 + 5 =$
Checked answer: $5 + 8$

What the question is asking

This question asks you to rewrite $8 + 5$ by swapping the order of the two numbers being added.

Important words

- The *commutative property of addition* says $a + b = b + a$. Order can switch; the sum stays the same.
- An *addend* is a number being added.

Work the problem one tiny step at a time

1. Start with $8 + 5$.
2. Switch the order of the two addends.
3. Write $5 + 8$.
4. Check: $8 + 5 = 13$ and $5 + 8 = 13$. Same sum.

Final answer: $5 + 8$

Why this makes sense

Commutative means "order can move." Do not change the operation to multiplication; keep addition.

K10

Question

Commutative: $7 \times a =$
Checked answer: $a \times 7$

What the question is asking

This question asks you to rewrite $7 \times a$ by swapping the order of the two factors being multiplied.

Important words

- The *commutative property of multiplication* says $a \times b = b \times a$. You may switch the order of the factors; the product stays the same.
- A *factor* is a number (or variable) being multiplied.
- The *associative property* only moves parentheses (regroups). It does **not** switch order. Commutative switches order.

Work the problem one tiny step at a time

1. Start with $7 \times a$.
2. Name the factors in order: first factor 7, second factor a.
3. Switch the order of the factors.
4. After the switch, first factor is a, second factor is 7: write $a \times 7$.
5. Numeric check with a number for a: if $a = 3$, then $7 \times 3 = 21$ and $3 \times 7 = 21$. Same product after the order switch.

Final answer: $a \times 7$

Why this makes sense

The product is the same either way. This is not the associative property (associative only regroups parentheses; commutative switches order).

K11

Question

Associative: $(b + 3) + 9 =$
Checked answer: $b + (3 + 9)$

What the question is asking

This question asks you to rewrite $(b+3)+9$ by moving the curved marks to group the addition in a different way.

Important words

- The *associative property of addition* says $(a + b) + c = a + (b + c)$. The grouping can change; the sum stays the same.
- *Regroup* means move the parentheses without changing the order of the terms.
- An *addend* is a number (or letter) being added.
- The *commutative property* switches order. Associative only moves parentheses; it does not reorder terms.

Work the problem one tiny step at a time

1. Start with $(b + 3) + 9$. Here b and 3 are grouped first.
2. List the three addends in order: b, then 3, then 9. Keep that order.
3. Move the parentheses to group 3 and 9: $b + (3 + 9)$.
4. Numeric check with a number for b: if $b = 2$, then $(2 + 3) + 9 = 5 + 9 = 14$.
5. Same value after regrouping: $2 + (3 + 9) = 2 + 12 = 14$. Check works.

Final answer: $b + (3 + 9)$

Why this makes sense

Associative changes grouping, not order. Writing $9+(b+3)$ would also equal the same value, but the standard associative rewrite keeps order and only moves parentheses. The numeric check $14 = 14$ confirms the regroup.

L. Distributive Property And Factoring

L1

Question

Factor: $8x + 16 =$
Checked answer: $8(x + 2)$

What the question is asking

This question asks you to write $8x + 16$ as a product by pulling out a shared whole number that divides both parts evenly.

Important words

- To *factor* means write a sum as a product (pull out a shared factor).
- The *GCF* of the terms is the largest factor shared by both terms.
- *Factored form* means written as a product, like $8(x + 2)$.
- Factoring undoes the distributive property: $a(b + c) = ab + ac$.

Work the problem one tiny step at a time

1. Look at the terms: $8x$ (coefficient 8) and the constant 16.
2. GCF of 8 and 16 is 8. (Variable x is only in the first term, so it is not part of the GCF.)
3. Divide first term by 8: $8x \div 8 = x$.
4. Divide second term by 8: $16 \div 8 = 2$.
5. Write the factored form: $8(x + 2)$.
6. Check by distributing: $8 \times x + 8 \times 2 = 8x + 16$. Matches.

Final answer: $8(x + 2)$

Why this makes sense

Always check by distributing back. If you get the original expression, the factoring is correct.

M. Decimal Computation

M1

Question

$428.5 + 37.26 =$ ___________ (decimal)
Checked answer: 465.76

What the question is asking

This question asks you to add 428.5 and 37.26 (numbers that use a dot for parts of a whole).

Important words

- A *decimal* uses a decimal point to show tenths and hundredths.
- To *add decimals*, line up the decimal points so equal place values are in the same column.
- The *sum* is the answer to an addition problem.

Work the problem one tiny step at a time

1. Write 428.5 as 428.50 to match hundredths.
2. Stack the numbers with decimal points lined up (see the grid below).
3. Add hundredths: $0 + 6 = 6$.
4. Add tenths: $5 + 2 = 7$.
5. Add ones: $8 + 7 = 15$. Write 5 in ones and carry 1 to the tens.
6. Add tens: $2 + 3 + 1$ (carry) $= 6$.
7. Add hundreds: $4 + 0 = 4$.
8. Sum: 465.76.

$$\begin{array}{r} 428.50 \\ +\ \ 37.26 \\ \hline 465.76 \end{array}$$

Final answer: 465.76

Why this makes sense

Estimate: $429 + 37 = 466$, close to 465.76.

M2

Question

$700.32 - 84.67 =$ ____________ (decimal)
Checked answer: 615.65

What the question is asking

This question asks you to subtract 84.67 from 700.32 and write the answer with a dot for parts of a whole.

Important words

- A *decimal* uses a decimal point to show tenths and hundredths.
- To *subtract decimals*, line up the decimal points so equal place values are in the same column.
- The *difference* is the answer to a subtraction problem.
- *Break apart* means split the number being subtracted into place-value parts and subtract one part at a time.

Work the problem one tiny step at a time

1. Both numbers already show two decimal places, so nothing needs padding: 700.32 and 84.67.
2. Stack 700.32 above 84.67 with the decimal points lined up (see the grid below). 84.67 has no hundreds digit, so its 8 sits in the tens column.
3. Break 84.67 into parts: $80 + 4 + 0.67$. Subtract one part at a time from 700.32.
4. Subtract the tens: $700.32 - 80 = 620.32$.
5. Subtract the ones: $620.32 - 4 = 616.32$.
6. Subtract six tenths: $616.32 - 0.60 = 615.72$.
7. Subtract seven hundredths: $615.72 - 0.07 = 615.65$.
8. So the difference is 615.65.

$$\begin{array}{r} 700.32 \\ -\quad 84.67 \\ \hline 615.65 \end{array}$$

Final answer: 615.65

Why this makes sense

Check by adding the difference to what you subtracted: $615.65 + 84.67$. Hundredths: $5 + 7 = 12$ (write 2, carry 1). Tenths: $6 + 6 + 1 = 13$ (write 3, carry 1). Ones: $5 + 4 + 1 = 10$ (write 0, carry 1). Tens: $1 + 8 + 1 = 10$ (write 0, carry 1). Hundreds: $6 + 0 + 1 = 7$. Result 700.32, which matches the starting number. Check passes.

M3

Question

$18.4 \times 12 =$ ___________ (decimal)
Checked answer: 220.8

What the question is asking

This question asks you to multiply 18.4 by 12.

Important words

- To *multiply decimals*: first multiply as if the numbers were whole numbers.
- Then place the decimal point by counting all digits after the decimal points in the factors.
- A *decimal digit* is a digit to the right of the decimal point.
- A *factor* is a number being multiplied. Here the factors are 18.4 and 12.
- The *product* is the answer to a multiplication problem.

Work the problem one tiny step at a time

1. Ignore the point: multiply 184×12.
2. $184 \times 10 = 1840$.
3. $184 \times 2 = 368$.
4. Add: $1840 + 368 = 2208$.
5. Count decimal digits: 18.4 has 1 decimal digit; 12 has 0 decimal digits; total 1 decimal place.
6. Place the point one digit from the right: 220.8.

Final answer: 220.8

Why this makes sense

Estimate: $18 \times 12 = 216$, close to 220.8.

M4

Question

$168.8 \div 8 =$ ___________ (decimal)
Checked answer: 21.1

What the question is asking

This question asks you to divide 168.8 by 8.

Important words

- The *dividend* is the number being divided (168.8).
- The *divisor* is the number you divide by (8).
- The *quotient* is the answer to a division problem.
- When dividing a decimal by a whole number, divide as usual and keep the decimal point in the same place in the quotient (line it up under the dividend's decimal point).
- To *bring down* a digit means move the next digit into the current leftover so you can keep dividing.

Work the problem one tiny step at a time

1. Divide 8 into 16: $8 \times 2 = 16$. Write 2 in the tens place of the quotient.
2. Bring down 8: $8 \div 8 = 1$. Write 1 in the ones place.
3. Place the decimal point in the quotient directly above the decimal point in 168.8.
4. Bring down 8 (tenths): $8 \div 8 = 1$. Write 1 in the tenths place.
5. Quotient: 21.1.
6. Check step: $21.1 \times 8 = 168.8$ (because $21 \times 8 = 168$ and $0.1 \times 8 = 0.8$).

$$\begin{array}{r} 2\,1\,.\,1 \\ 8\;\overline{)\,1\,6\,8\,.\,8} \end{array}$$

Quotient digits sit above matching places; the decimal points line up.

Final answer: 21.1

Why this makes sense

Check: 21.1×8. $21 \times 8 = 168$ and $0.1 \times 8 = 0.8$, total 168.8. Correct.

M5

Question

$4.2 \times 1.5 =$ ___________ (decimal)

Checked answer: 6.3

What the question is asking

This question asks you to multiply 4.2 by 1.5.

Important words

- A *factor* is a number being multiplied. Here the factors are 4.2 and 1.5.
- The *product* is the answer to a multiplication problem.
- To *multiply decimals*: multiply as if they were whole numbers, then place the decimal point by counting all decimal digits in the factors.
- A *decimal digit* is a digit to the right of the decimal point.

Work the problem one tiny step at a time

1. Ignore the points for a moment: multiply 42×15.
2. $42 \times 10 = 420$.
3. $42 \times 5 = 210$.
4. Add: $420 + 210 = 630$.
5. Count decimal digits: 4.2 has 1, 1.5 has 1, total 2.
6. Place the point two places from the right in 630: 6.30.
7. Drop the trailing zero: $6.30 = 6.3$.

Final answer: 6.3

Why this makes sense

Estimate: $4 \times 1.5 = 6$, close to 6.3.

M6

Question
$7.2 \div 0.8 =$ ___________ (decimal)
Checked answer: 9

What the question is asking
This question asks you to divide 7.2 by 0.8.

Important words

- The *dividend* is the number being divided (7.2).
- The *divisor* is the number you divide by (0.8).
- The *quotient* is the answer to a division problem.
- A *power of ten* is a number like 10, 100, or 1000 made by multiplying tens together.
- To *divide by a decimal*, multiply the dividend and the divisor by the same power of ten so the divisor becomes a whole number.

Work the problem one tiny step at a time

1. Name the parts: dividend 7.2, divisor 0.8.
2. Multiply both by 10 (one power of ten) to clear one decimal place in the divisor: $7.2 \times 10 = 72$ and $0.8 \times 10 = 8$.
3. Now compute $72 \div 8$.
4. $8 \times 9 = 72$, so $72 \div 8 = 9$.
5. Therefore $7.2 \div 0.8 = 9$.
6. Check: $0.8 \times 9 = 7.2$.

Final answer: 9

Why this makes sense
Check: $0.8 \times 9 = 7.2$. Yes.

N. Fraction Operations And Properties

N1

Question
$\frac{1}{2} + \frac{1}{4} =$ ___________ (simplest)
Checked answer: $\frac{3}{4}$

What the question is asking

This question asks you to add $\frac{1}{2}$ and $\frac{1}{4}$, then shrink the answer if something bigger than 1 divides both the top and the bottom evenly.

Important words

- A *common denominator* is a shared bottom number so fractions can be added.
- To add fractions with the same denominator, add the numerators and keep the denominator.

Work the problem one tiny step at a time

1. The denominators are 2 and 4. A common denominator is 4.
2. Change $\frac{1}{2}$ into fourths: multiply top and bottom by 2: $\frac{1 \times 2}{2 \times 2} = \frac{2}{4}$.
3. Now add: $\frac{2}{4} + \frac{1}{4} = \frac{2+1}{4} = \frac{3}{4}$.
4. $\frac{3}{4}$ is already in simplest terms.

Final answer: $\frac{3}{4}$

Why this makes sense

One half is two fourths. Two fourths plus one fourth is three fourths.

N2

Question

$\frac{5}{6} - \frac{1}{3} =$ ____________ (simplest)

Checked answer: $\frac{1}{2}$

What the question is asking

This question asks you to subtract $\frac{1}{3}$ from $\frac{5}{6}$, then shrink the answer if needed.

Important words

- A *common denominator* is a shared bottom number so fractions can be subtracted.
- To *simplify* a fraction means divide top and bottom by the same common factor until they share no common factor greater than 1.
- To subtract fractions with the same denominator, subtract the numerators and keep the denominator.

Work the problem one tiny step at a time

1. Denominators 6 and 3. Common denominator: 6.
2. Change $\frac{1}{3}$: multiply top and bottom by 2: $\frac{2}{6}$.

3. Subtract: $\frac{5}{6} - \frac{2}{6} = \frac{3}{6}$.
4. Simplify $\frac{3}{6}$: write the division facts $3 \div 3 = 1$ and $6 \div 3 = 2$.
5. So $\frac{3}{6} = \frac{1}{2}$.

Final answer: $\frac{1}{2}$

Why this makes sense

Five sixths take away two sixths leaves three sixths, which is one half.

N3

Question

$\frac{5}{8} \times \frac{4}{15} =$ ___________ (simplest)

Checked answer: $\frac{1}{6}$

What the question is asking

This question asks you to multiply the equal-parts-over-a-whole amounts and shrink the answer until nothing bigger than 1 divides both the top and the bottom evenly.

Important words

- To *multiply fractions*: multiply numerators; multiply denominators. Then simplify.
- To *cancel* means divide out the same factor from a numerator and a denominator before multiplying.

Work the problem one tiny step at a time

1. Write $\frac{5 \times 4}{8 \times 15} = \frac{20}{120}$.
2. Simplify $\frac{20}{120}$: write $20 \div 20 = 1$ and $120 \div 20 = 6$, so $\frac{1}{6}$.
3. Or cancel first: 4 and 8 share a factor of 4, so divide those out: $\frac{5}{8} \times \frac{4}{15} = \frac{5}{2} \times \frac{1}{15}$.
4. Multiply: $\frac{5 \times 1}{2 \times 15} = \frac{5}{30}$.
5. Also cancel: 5 and 30 share a factor of 5: $\frac{5}{30} = \frac{1}{6}$ (because $5 \div 5 = 1$ and $30 \div 5 = 6$).

Final answer: $\frac{1}{6}$

Why this makes sense

Cross-check: $5 \times 4 = 20$ and $8 \times 15 = 120$, and $\frac{20}{120} = \frac{1}{6}$.

N4

Question
$\frac{7}{8} \div \frac{1}{4} =$ ____________ (simplest)
Checked answer: $\frac{7}{2}$

What the question is asking

This question asks you to divide $\frac{7}{8}$ by $\frac{1}{4}$, then shrink the answer if needed.

Important words

- The *reciprocal* of a fraction flips the numerator and the denominator (swap top↔bottom). Example: the reciprocal of $\frac{1}{4}$ is $\frac{4}{1}$.
- To *flip* a fraction means the same swap: top↔bottom.
- To *divide by a fraction*, multiply by its reciprocal.
- To *cancel* means divide out the same factor from a numerator and a denominator before multiplying (same idea as N3).

Work the problem one tiny step at a time

1. Rewrite: $\frac{7}{8} \div \frac{1}{4} = \frac{7}{8} \times \frac{4}{1}$.
2. Multiply: $\frac{7 \times 4}{8 \times 1} = \frac{28}{8}$.
3. Simplify $\frac{28}{8}$: write $28 \div 4 = 7$ and $8 \div 4 = 2$, so $\frac{7}{2}$.
4. Or cancel 4 with 8 first (same factor top and bottom): $\frac{7}{8} \times \frac{4}{1} = \frac{7}{2}$.

Final answer: $\frac{7}{2}$

Why this makes sense

How many fourths fit into seven eighths? Multiplying by 4 scales the amount: $\frac{7}{2} = 3\frac{1}{2}$.

N5

Question
Commutative: $7 \times a =$ ____________
Checked answer: $a \times 7$

What the question is asking

This question asks you to rewrite $7 \times a$ by swapping the order of the two factors being multiplied (order can change; the answer stays the same).

Important words

- The *commutative property of multiplication* says $a \times b = b \times a$. You may switch the order of the factors; the product stays the same.
- A *factor* is a number or letter being multiplied.

Work the problem one tiny step at a time

1. Start with $7 \times a$.
2. Name the factors in order: first factor 7, second factor a.
3. The commutative property lets you switch the order of the factors.
4. After the switch, first factor is a, second factor is 7: write $a \times 7$.
5. Numeric check with a number for a: if $a = 3$, then $7 \times 3 = 21$ and $3 \times 7 = 21$. Same product.

Final answer: $a \times 7$

Why this makes sense

Order changes; the product meaning stays the same. This is not the associative property (associative only moves parentheses).

N6

Question

Associative: $(x + 5) + 8 =$ ____________
Checked answer: $x + (5 + 8)$

What the question is asking

This question asks you to rewrite $(x + 5) + 8$ by moving the curved marks to group the addition in a different way (grouping can change; the answer stays the same).

Important words

- The *associative property of addition* says $(a+b)+c = a+(b+c)$. You may change the grouping; the sum stays the same.
- *Regroup* means move the parentheses without changing the left-to-right order of the terms.

Work the problem one tiny step at a time

1. Start with $(x + 5) + 8$. Here x and 5 are grouped first.
2. List the three addends in order: x, then 5, then 8. Keep that order.
3. Associative moves only the parentheses. Do not reorder into $8 + (x + 5)$.
4. Move the parentheses to group 5 and 8: $x + (5 + 8)$.
5. Numeric check with a number for x: if $x = 2$, then $(2 + 5) + 8 = 7 + 8 = 15$ and $2 + (5 + 8) = 2 + 13 = 15$. Same sum.

Final answer: $x + (5 + 8)$

Why this makes sense

Associative regroups; it does not reorder. Writing $8 + (x + 5)$ equals the same sum, but the required associative rewrite keeps order and only moves parentheses.

N7

Question

Is subtraction associative? Compare $(12 - 4) - 3$ and $12 - (4 - 3)$: ___________
Checked answer: No

What the question is asking

This question asks whether subtraction still gives the same answer when you group the numbers in two different ways.

Important words

- *Associative* would mean regrouping never changes the answer.
- If the two results are different, subtraction is *not* associative.

Work the problem one tiny step at a time

1. Compute the left grouping: $(12 - 4) - 3$.
2. Inside first: $12 - 4 = 8$.
3. Then $8 - 3 = 5$.
4. Compute the right grouping: $12 - (4 - 3)$.
5. Inside first: $4 - 3 = 1$.
6. Then $12 - 1 = 11$.
7. Compare: $5 \neq 11$.
8. Because the answers differ, subtraction is not associative. Answer: No.

Final answer: No

Why this makes sense

Addition is associative, but subtraction is not. Changing parentheses in subtraction can change the result, as 5 versus 11 shows.